

SENIOR SECONDARY EXAMINATION, 2015

CHEMISTRY

TIME: 3:1/4

MARKS:56

GENERAL INSTRUCTION TO THE EXAMINATIONS:

1. Candidate must write first his / her Roll No. on the question paper compulsorily.
2. All the question are compulsory.
3. Write the answer each question in the given answer-book only.
4. For question having more than one part the answers to those parts are to be written together
5. If there is any error/ difference / contradiction in Hindi & English versions of the question paper, the question of Hindi version should be treated valid.

1. Write the formula to calculate the molarity. 1

Sol. The molarity is equal to the number of moles of solute divided by the volume of the solution measured in liters. In this equation X, Y, and Z represent 2, 6 and 3 for example.

2. Write the SI unit of Resistivity. 1

Sol. The SI unit of Resistivity is the Ohm-metre

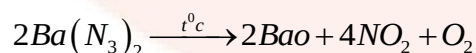
3. Limit molar conductivities of Mg^{2+} and Cl^{-} ions are $106.0 \text{ s cm}^2 \text{ mol}^{-1}$ and $76.3 \text{ cm}^2 \text{ mol}^{-1}$ respectively calculate the limiting molar conductivity $MgCl_2$. 1

Sol.

Limiting molar conductance of $MgCl_2$ = Limiting molar conductance of Mg^{2+} + 2(limiting molar conductance of Cl^{-} -ion) = $(106.05 + 2 \times 76.35) = 258.75$

4. What happens on thermal decomposition of $Ba(N_3O)_2$? Write only equation of the reaction. 1

Sol.



5. Write IUPAC name of $C_2H_5 - \underset{\substack{| \\ C_2H_5}}{N} - CH_2 - CH_2 - CH_3$. 1

Sol.

6. Write the formula of ionisation isomer of $[Cr(H_2O)_4Br_2]Cl$. 1

Sol. The formula of isonisation isomer of $[Cr(H_2O)_4Cl]Br_2$

Quadaqua chloro chromium iii Bromine.

7. Write the formula of coordination compound mercury tetrathiocyanato-cobaltate (iii). 1

Sol. $Hg[Co(SCN)_4]$

8. What happens when benzene diazonium chloride reacts with potassium iodide? (Write only equation of the reaction). 1

Sol. $C_6H_5-N^+Cl^- + KI \xrightarrow{\Delta} C_6H_5-I + N_2 \uparrow + KCl$

9. Explain that vulcanized rubber is an elastomer. 1

Sol.

Vulcanization of rubber is process of improvement of the rubber elasticity and strength by heating it in the presence of sulfur, which results in three-dimensional cross-linking of the chain rubber molecules bonded to each other by sulfur atoms.

10. For a reaction the unit of rate constant is sec^{-1} . What will be the order of reaction? 1

Sol.

It is a second order reaction, the units of k is given by

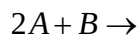
$$k = (molL^{-1})^{1-n} min^{-1}$$

$$k = (molL^{-1})^{1-2} min^{-1} = mol^{-1}L min^{-1}$$

11. Write differential rate equation for the reaction, $2A + B \rightarrow product$. 1

Sol.

The differential rate equation for the reaction



$$K[A]^0[B]^0$$

12. What is basicity of H_3PO_3 ? 1

Sol.

The basicity is determined by the number of H atoms attached directly to the O atom in the molecule.

13. Arrange benzoic acid, 4-methyl benzoic acid and 4-nitrobenzoic acid in increasing order of their acidic strength. 1

Sol.

The arrange increasing order of their acidic strength.

4-methyl benzoic acid < benzoic acid, < 4-nitrobenzoic acid

14. Explain calcination and roasting with example. 1+1=2

Sol.

The calcination process of subjective a substance to heat in order to cause some change in tis physical or chemical constitution. Calcination is usually done to:

- (1) To remove water, present as a absorbed moisture, as water of crystallization;
- (2) To remove carbon dioxide, sulfur dioxide, or other volatile constituent
- (3) To oxidize a part or the whole of the substance.

Example roasting: $2ZnS + 2O_2 \rightarrow 2ZnO + 2SO_2 \uparrow$

15. Explain denaturation of protein with an example. 1+1=2

Sol.

Denaturation of protein involves the disruption and possible destruction of both the secondary and tertiary structures. Since denaturation reactions are not strong enough to break the peptide bonds, the primary structure remains the same after a denaturation process. Denaturation disrupt the normal alpha-helix and beta sheet protein and uncoils it into a random shape.

16. Osmotic pressure of a solution is 0.0821 atm at a temperature of 400 k. calculate the concentration of solution in mol/litre. [R=0.0821 L atm K⁻¹ mol⁻¹]
 $\frac{1}{2} + 1 \frac{1}{2} = 2$

Sol. The osmotic pressure

$$\pi = CRT$$

$$0.0821 \text{ atm} = C \times 0.0821 \text{ L atm / K mol} \times 400 \text{ K}$$

$$C = 0.0025 \text{ mol / L}$$

Hence, the concentration of solution is 0.0025 mol/k.

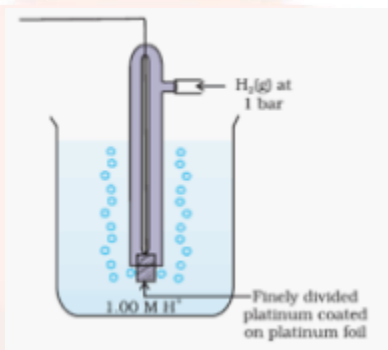
17. (a) Write Faraday's second of electrolysis.

Sol. Faraday's second: Faraday's second law of electrolysis states that when the same quantity of electricity is passed through several electrolytes, the mass of the substance deposited are proportional to their respective chemical equivalent or equivalent weight.

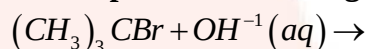
(b) Draw labelled diagram of standard Hydrogen electrode.

1+1=2

Sol.



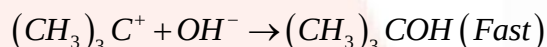
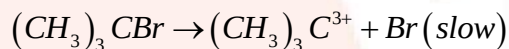
18. Complete the following reaction and explain its mechanism :



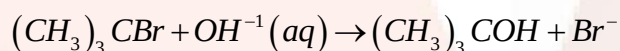
$\frac{1}{2} + 1 \frac{1}{2} = 2$

Sol.

Mechanism:



The below equation is complete reaction.



19. Arrange the following compounds in increasing order of their basic strength.

Explain reason:



1+1=2

Sol. Arrange the compounds in increasing order of their basis strength.



Always aliphatic amines are more basic than aromatic amines. This is because the availability of electron on nitrogen to donate is less in aromatic amines due to resonance.

In aliphatic amines more the number of alkyl group due to their +I effect, they donate electrons towards N and more electrons on nitrogen atom to donate.

20. Write any two differences between addition and condensation polymerization.
1+1=2

Sol.

The main difference between addition and condensation polymerization is that in addition polymerization the polymers are formed by the addition of monomers with no by-products while in condensation polymerization polymers are formed due the condensation of two different monomers results in the formation of small

21. (a) Write an equation of the reaction of chlorine with cold and dilute NaOH solution.

Sol.

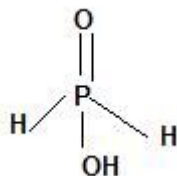


(b) Explain the reducing nature of H_3PO_2 .

1+1=2

Sol.

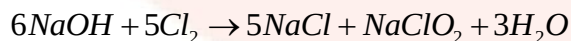
There are 2 P-H bonds which makes it acidic too, and it can provide hydrogen and reduce the other species So it will be the reducing agent.



OR

(a) Write an equation of reaction of chlorine with hot and concentrated NaOH solution.

Sol.



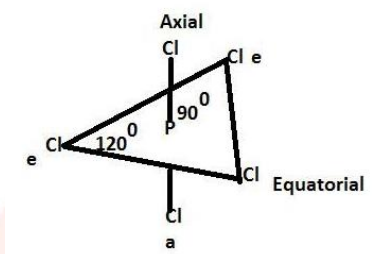
(b) Why are all the five bonds of PCl_5 not equivalent? Explain.

1+1=2

Sol.

PCl_5 has a triangle bipyramidal structure due to the dsp^3 hybridization. Due to this trigonal bipyramidal structure, the two Cl-atoms of PCl_5 molecule are stretched at the axis and become

little longer and so called as axial bond, while the rest three Cl-atoms lie on the equator and called as equatorial bonds. The axial bonds are more stretched and longer than equatorial bonds so, the three equatorial Cl-atoms become equivalent and the two axial Cl-atom become equivalent while, both the groups are not identical which makes the difference in bonding of PCl_5 molecule as follows:



22. Half-life period for first order reaction is 10 sec. calculate the rate constant for the reaction.

$$\frac{1}{2} + 1 \frac{1}{2} = 2.$$

Sol.

We have,

First order reaction of 10 Sec.

Therefore,

According to the formula of first order reaction we get,

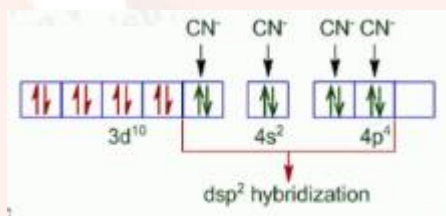
$$\begin{aligned} t_{1/2} &= \frac{\ln 2}{k} \\ &= \frac{0.693}{k} \\ \Rightarrow k &= \frac{0.693}{10} \\ &= 0.0693s^{-1} \end{aligned}$$

Hence, the rate constant for the reaction $0.0693s^{-1}$

23. Explain on the basis of valence bond theory that $[Ni(CN)_4]^{2-}$ is a low spin complex ion. 2

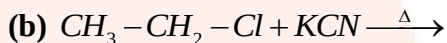
Sol.

The basic of valence bond theory. The 4d, 3s and two 4p orbitals undergo dsp^2 -hybridization to make bond with CN^- ligands in square planar geometry. Thus $[Ni(CN)_4]^{2-}$ is diamagnetic.

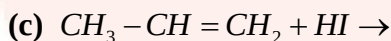
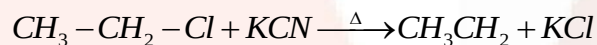


24. Complete the following chemical reactions:

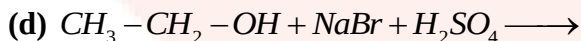
(a)



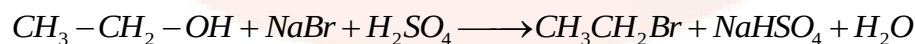
Sol.



Sol.



Sol.



25. (a) Write the value of axial angles for hexagonal crystal system.

Sol.

The monoclinic class has unequal axial length $a \neq b \neq c$ but the axial angles are $\alpha = \gamma = 90^\circ \neq \beta$. The angle β lies between a and c axes and it is obtuse between 90° to 180° .

(b) Which type of semiconductor is obtained by doping boron with silicon? Explain.
1+2=3

Sol.

In semiconductor production, doping intentionally introduces impurities into an extremely pure intrinsic semiconductor for the purpose of modulating its electrical properties. The impurities are dependent upon the type of semiconductor and properties that it needs to have for its intended purpose. Lightly and moderately doped semiconductors are referred to as extrinsic semiconductors. A semiconductor doped to such high levels that it acts more like a conductor than a semiconductor is referred to as a degenerate semiconductor.

1+1+1=3

26. (a) What is lanthanoid contraction

Sol.

Lanthanoid contraction is the decrease in size of lanthanoids from left to right due to the increase in effective nuclear charge. This increase in effective nuclear charge is due to the poor shielding effect of f-subshell. The consequence of lanthanoid contraction is observed on normal group elements also. For Zr and Hf have similar atomic sizes because of lanthanoid contraction. As Hf is down to Zr in the same group, so size should increase, but due to lanthanoid contraction it decreases and is almost similar to Zr and therefore properties are also similar.

(b) What are interstitial compounds? Give an example.

Sol.

An interstitial compound or interstitial alloy is a compound that is formed when an atom of sufficiently small radius sits in an interstitial "hole" in a metal lattice. Example – small atoms are hydrogen, boron, carbon and nitrogen.

(c) Calculate the 'Spin only' magnetic moment of M^{2+} (aq) ion.
(Z = 27)

Sol.

$$Z = 27$$

$\Rightarrow$

$$[M^{2+}] = [Ar]3d^7$$

$$3d^7 =$$



3 unpaired electrons

$$N=3$$

$$\Rightarrow \sqrt{n(n+3)} = \mu$$

$$\Rightarrow \sqrt{3(3+3)} = \mu$$

$$\Rightarrow \sqrt{15} = \mu$$

$$\Rightarrow 4BM$$

27.(a) What are non-narcotic analgesics? Write an example.

$$\frac{1}{2} + 1 + \frac{1}{2} + 1 + 1 = 3$$

Sol. Non-narcotic analgesics are medications used to control pain and inflammation. They are available at drugstores without a prescription or by prescription when given at higher doses. Some type of these medications can be given during surgery to reduce post-surgical pain and lessen the need for narcotics.

Example: Tylenol (chemical name: acetaminophen);

(b) Why is artificial sweetening agent, Aspartame used only in soft drinks and cold foods?

Sol. Aspartame is better choice for diabetics as it is a non-saccharide sweetener. Aspartame is a methyl ester of the aspartic acid and phenylalanine dipeptide.

(c) Write any one difference between antiseptics and disinfectants.

Sol. The difference between antiseptics and disinfectants are both disease causing organisms. The difference is how each substance is used. Antiseptics are applied to living skin or tissue to prevent infection, whereas disinfectants are applied to surface, equipment or other inanimate object.

OR,

$$\frac{1}{2} + \frac{1}{2} + 1 + 1 = 3$$

(a) What is ionic detergent? Write an example.

Sol. Examples include octyl thioglucoside and maltoside HEGA and MEGA series detergents are similar, possessing a sugar alcohol as headgroup. Zwitterionic detergents possess a net zero charge arising from the presence of equal number of +1 and -1 charged chemical group.

Examples includes CHAPS.

(b) Why are chemicals added to food? Write any two reasons.

Sol. Food added in chemical because food additives are substance added to food to preserve flavor or enhance its taste, appearance, or other qualities. Some additives have been used for centuries for example, preserving food by pickling, salting, as with bacon, preserving sweets or using sulfur dioxide as with wines.

(c) Write any one difference between broad spectrum and narrow spectrum antibiotics.

Sol. Broad spectrum antibiotics act against multiple strains and forms of different bacteria which share common structures and metabolic functions that can be attached and affected to kill them. Narrow spectrum antibiotics are more specific in their course and act against only certain bacteria as a more targeted approach.

28. (a) What is enzyme catalysis? Write an example.

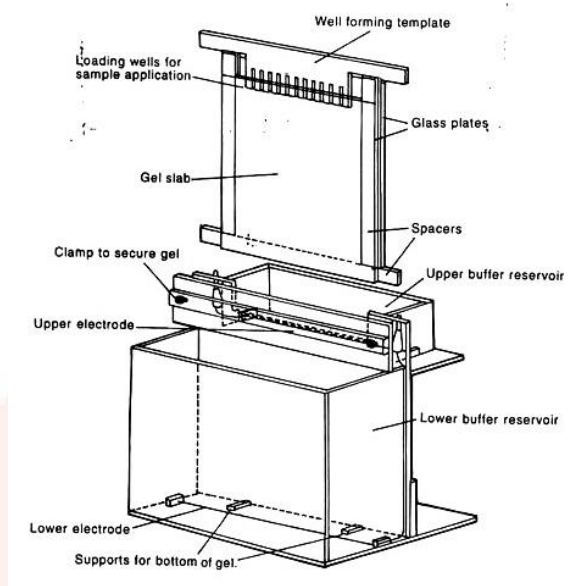
Sol. Enzyme Catalysts are biological and chemical catalysts which are for example in our stomach, The catalysts are generally +ve in nature they provide an increase in the reaction and so called as activator. But when the catalysts reduce the rate of reaction, they are called as inhibitors or -ve catalyst. When, some catalysts are called 'outo catalysts' in which some of the products work as catalyst for the reaction.

(b) Which type of emulsion is milk? Explain.

Sol. Milk is an example of the oil-in-water type of emulsion. In milk liquid fat globules are dispersed in water. The emulsion in which water forms the dispersed phase, and the oil acts as the dispersion medium is called a water – in – oil emulsion.

(d) Explain Electrophoresis with labelled diagram.

Sol.



OR,

$$\frac{1}{2} + \frac{1}{2} + 1 + 1 = 4$$

(a) What is selectivity for catalyst. Give for example.

Sol.

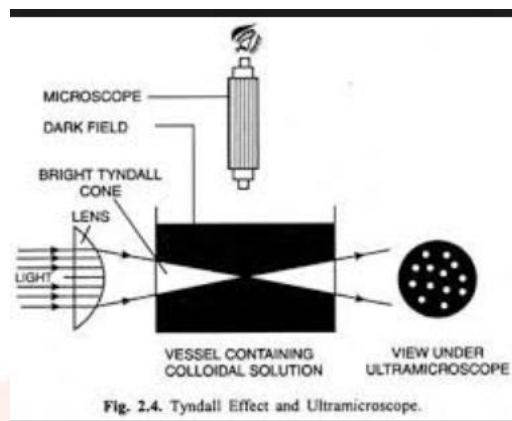
Selectivity of Catalyst. Catalysts are highly specific compounds. They have an ability to direct the reaction to yield a particular product. The reaction with same reactants but different catalyst may yield different products. This is termed as the selectivity of catalyst.

(b) What happens when hydrated ferric oxide sol is mixed with arsenium Isulphide sol?

Sol. when hydrated ferric oxide sol is mixed with arsenium Isulphide sol then it they neutralize each other being oppositely charged.

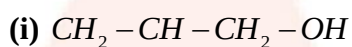
(c) Explain Tyndall effect with labelled diagram.

Sol. Passing through a colloidal mixture, get scattered by its particles. The visible beam of headlights in fog is caused by the Tyndall effect. The water droplets scatter the light, making the headlight beam visible. Is called tyndall effect.



29. Write IUPAC names of the following compound:

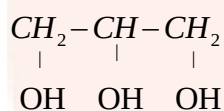
1+1+1+1=4



Sol.

$\text{CH}_2 - \text{CH} - \text{CH}_2 - \text{OH}$ IUPAC name is 1,2- enenediaol.

(ii)

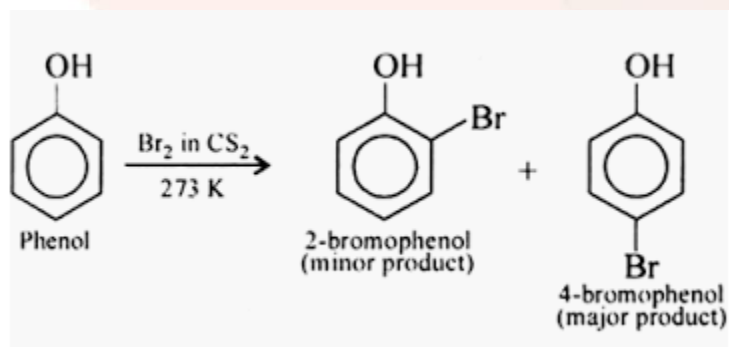


Sol. 1,2,3 - propanetriaol

(b) Write equation of the following chemical reactions:

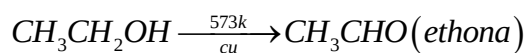
(i) Phenol reacts with Br_2 in presence of CS_2 .

Sol.



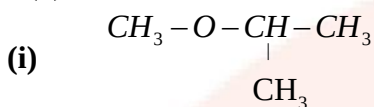
(ii) Ethanol is heated at 573 K in presence of Cu.

Sol.

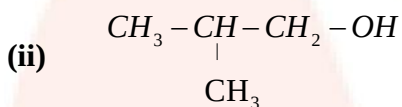


OR,

(a) Write IUPAC of the following compounds:



Sol. 2-methyl-1-propanone.



Sol:

2-methyl propan-1-ol

(b) Write equation for chemical reaction.

(i) Phenol reacts with conc. HNO_3 .

Sol.

Phenol reacts with conc HNO_3 . When phenol reacts with nitric acid at room temperature, electrophilic substitution occurs to form a mixture of 2-nitrophenol and 4-nitrophenol. This increased substitution is due to concentrated nitric acid having a greater amount of NO_2^+ nucleophile per unit volume.

(ii) Phenol reacts with zinc dust.

Sol..

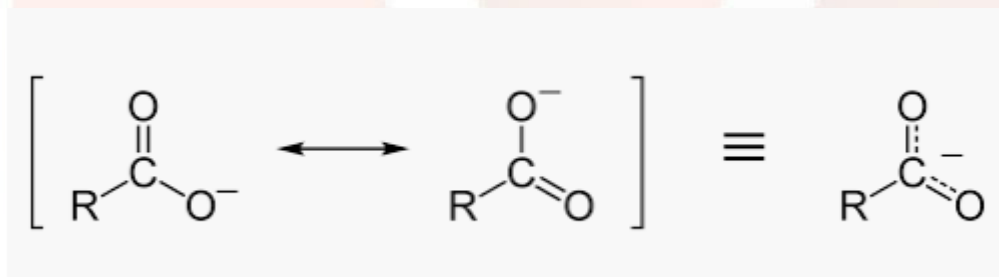


30. (a) Write a note on Resenmund reduction.**Sol.**

The reason, a hydrogenolysis, is catalyzed by palladium on barium sulfate, which is sometimes called the Rosenmund catalyst. Barium sulfur has a low surface area which reduces the activity of the palladium, preventing over-reduction. However for certain reactive acylchlorides the activity must be reduced further, by the addition of a Polson. Originally this was thioquinathrene although thiourea has also been used. Deactivation is required because the system must reduce the acyl chloride but not the subsequent aldehyde. If further reduction does take place it will create a primary alcohol which would then react with the remaining acyl chloride to form ester.

(b) Why is fluoroacetic acid more acidic than chloroacetic acid?

Sol. Because the presence of fluoride increases its acidity as fluorine is acidic in nature but on the other hand to acetic acid there is no other atom joined.

(c) Draw the resonating structure of carboxylate ion.**Sol.****OR,****(a) Write a note on Wolff–Kishner reduction.**

Sol. Wolff-Kishner reduction is an organic reaction used to convert an aldehyde or an alkane using hydrazine, base and thermal conditions. The mechanism begins with the attack of hydrazine on the aldehyde or ketone to form an imine.

(b) Boiling point of carboxylic acids are higher than aldehydes and ketones of comparable molecular masses why?**Sol.**

Carboxylic acids are higher boiling liquids than aldehydes, ketones and even alcohols of comparable molecular masses. This is due to extensive association of carboxylic acid molecules through intermolecular hydrogen bonding.

(C) Draw the dimer structure of ethanoic acid in vapour state.

Sol.

